# Masked observations

## 1 Masked observations

**Theorem 1.1.** *For every natural number $t$,* $\mathrm{whitecount}(t, 0) = 0$.

*Proof.* The goal follows from $\mathrm{whitecountzero}(t)$. □

**Theorem 1.2.** *For every natural number $t$ and natural number $n$, if* $\mathrm{center}(t) = \mathrm{true}$*, then* $\mathrm{whitecount}(t, n + 1) = \mathrm{whitecount}(t + 1, n)$.

*Proof.* The goal follows from $\mathrm{whitecountblackstep}(t, n)$. □

**Theorem 1.3.** *For every natural number $t$ and natural number $n$, if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{whitecount}(t, n + 1) = \mathrm{whitecount}(t + 1, n) + 1$.

*Proof.* The goal follows from $\mathrm{whitecountwhitestep}(t, n)$. □

**Theorem 1.4.** *For every natural number $t1$ and natural number $t2$ and natural number $n$, if* $\mathrm{centerwindowagree}(t1, t2, n)$*, then* $\mathrm{leftprefixagree}(t1, t2, n)$ *if and only if* $\mathrm{whitetraceagree}(t1, t2, n)$.

*Proof.* The goal follows from $\mathrm{prefixagreementiffwhitetrace}(t1, t2, n)$. □

**Theorem 1.5.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then there exists a natural number $a$ such that there exists a natural number $b$ such that $a < b$ and $b \leq$* $\mathrm{twopow}(\mathrm{whitecount}(t, n)) \wedge \mathrm{leftprefixagree}(t + a \cdot P, t + b \cdot P, n)$.

*Proof.* The goal follows from $\mathrm{compressedprefixrepeat}(T, P, t, n)$. □

**Theorem 1.6.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$ and natural number $M$, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then if* $\mathrm{prefixseparated}(t, P, n, M)$*, then* $M \leq \mathrm{twopow}(\mathrm{whitecount}(t, n))$.

*Proof.* The goal follows from $\mathrm{separatedprefixbound}(T, P, t, n, M)$. □

**Theorem 1.7.** *For every natural number $T$ and natural number $P$ and natural number $t$ and natural number $n$, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq t$, then* $n < t + \mathrm{twopow}(\mathrm{whitecount}(t, n)) \cdot P$.

*Proof.* The goal follows from $\mathrm{edgeinformationbound}(T, P, t, n)$. □

**Theorem 1.8.** *For every natural number $n$ and natural number $t$ and natural number $w$ and natural number $P$, if $n < (w+1) \cdot P$, then $n < t + \mathrm{twopow}(w) \cdot P$.*

*Proof.* The goal follows from $\mathrm{informationboundaudit}(n, t, w, P)$. □

## 2 Central collision recurrence

**Theorem 2.1.** *For every natural number $Q$ and natural number $t$ and natural number $r$, if $\mathrm{leftadjacentdefect}(Q, t) = \mathrm{true}$, then if $\mathrm{center}(t + r) = \mathrm{true}$, then if $\mathrm{center}(t + r + Q) = \mathrm{true}$, then $\mathrm{nearcollision}(Q, t, r + 1, 2 \cdot (r + 1))$.*

*Proof.* The goal follows from $\mathrm{unitdefectforcesnearcollision}(Q, t, r)$. □

**Theorem 2.2.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then if $T \leq t$, then if $\mathrm{leftadjacentdefect}(m \cdot P, t) = \mathrm{true}$, then $\mathrm{nearcollision}(m \cdot P, t, P, 2 \cdot P)$.*

*Proof.* The goal follows from $\mathrm{periodicunitdefectforcesnearcollision}(T, P, m, t)$. □

**Theorem 2.3.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + P \wedge \mathrm{centralcollision}(m \cdot P, t, 2 \cdot P)$.*

*Proof.* The goal follows from $\mathrm{periodiccentralcollisionwindow}(T, P, m, N)$. □

**Theorem 2.4.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then there exists a natural number $t$ such that $N \leq t$ and $\mathrm{centralcollision}(m \cdot P, t, 2 \cdot P)$.*

*Proof.* The goal follows from $\mathrm{periodiccentralcollisions}(T, P, m, N)$. □

**Theorem 2.5.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if $\mathrm{centerperiodic}(T, P)$, then $\mathrm{recurrentcentralcollision}(m \cdot P, 2 \cdot P)$.*

*Proof.* The goal follows from $\mathrm{periodicrecurrentcentralcolumn}(T, P, m)$. □

**Theorem 2.6.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b+r) =$ true*, then there exists a natural number $s$ such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*leftcollision$(Q, b+s, r+1)$.

*Proof.* The goal follows from blackwindowforcesleftcollision$(Q, b, r)$. □

**Theorem 2.7.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* leftcollision$(m \cdot P, t, P)$.

*Proof.* The goal follows from periodicleftcollisionwindow$(T, P, m, N)$. □

**Theorem 2.8.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then there exists a natural number $t$ such that $N \leq t$ and* leftcollision$(m \cdot P, t, P)$.

*Proof.* The goal follows from periodicleftcollisions$(T, P, m, N)$. □

**Theorem 2.9.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentleftcollision$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentleftcolumn$(T, P, m)$. □

**Theorem 2.10.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* collisioncode$(Q, t, k) \neq 0$ *if and only if* collisionbit$(Q, t, k) =$ true.

*Proof.* The goal follows from collisioncodenonzero$(Q, t, k)$. □

**Theorem 2.11.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* collisioncode$(Q, t, k) \leq 6$.

*Proof.* The goal follows from collisioncodebound$(Q, t, k)$. □

**Theorem 2.12.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then* collisioncode$(Q, t, k) = 1 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 2$ *and* (collisioncode$(Q, t, k) = 2 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 3 \leftrightarrow$ leftcolour$(Q, t, k) =$ $1 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 4 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$leftcolour$(Q, t, k-1) = 1) \wedge$(collisioncode$(Q, t, k) = 5 \leftrightarrow$ leftcolour$(Q, t, k) =$ $2 \wedge$leftcolour$(Q, t, k-1) = 3) \wedge$(collisioncode$(Q, t, k) = 6 \leftrightarrow$ leftcolour$(Q, t, k) =$ $3 \wedge$ leftcolour$(Q, t, k - 1) = 2)$.

*Proof.* The goal follows from collisioncodecolours$(Q, t, k)$. □

**Theorem 2.13.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* advancingatdepth$(Q, t, k)$ *if and only if* collisioncode$(Q, t, k) = 3$ *or* collisioncode$(Q, t, k) = 5$.

*Proof.* The goal follows from advancingcollisioncodes$(Q, t, k)$. □

**Theorem 2.14.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then* leftdefect$(Q, t + 1, k) =$ leftdefect$(Q, t, k + 1)$.

*Proof.* The goal follows from collisiontransmitsleft$(Q, t, k)$. □

**Theorem 2.15.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then* leftgatedifference$(Q, t, k) =$ false.

*Proof.* The goal follows from collisiongatesagree$(Q, t, k)$. □

**Theorem 2.16.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $2 \leq k$, then* collisioncode$(Q, t, k) = 1 \to$ leftdefect$(Q, t+1, k-1) =$ leftcell$(t, k-2)$ *and* (collisioncode$(Q, t, k) = 2 \to$ leftdefect$(Q, t+1, k-1) =$ leftcell$(t+Q, k-2)) \wedge$(collisioncode$(Q, t, k) = 3 \to$ leftdefect$(Q, t+1, k-1) =$ true$) \wedge$ (collisioncode$(Q, t, k) = 4 \to$ leftdefect$(Q, t + 1, k - 1) =$ not(leftcell$(t + Q, k - 2))) \wedge$ (collisioncode$(Q, t, k) = 5 \to$ leftdefect$(Q, t + 1, k - 1) =$ true$) \wedge$ (collisioncode$(Q, t, k) = 6 \to$ leftdefect$(Q, t + 1, k - 1) =$ not(leftcell$(t, k - 2))$).

*Proof.* The goal follows from collisionrighttransporttable$(Q, t, k)$. □

**Theorem 2.17.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then* collisioncode$(Q, t, k) = 3 \to$ leftcolour$(Q, t + 1, k-1) = 2$ *and if* collisioncode$(Q, t, k) = 5$*, then* leftcolour$(Q, t+1, k-1) =$ 1.

*Proof.* The goal follows from advancingrightcolourtable$(Q, t, k)$. □

**Theorem 2.18.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftdefect$(Q, t + 1, k) =$ not(leftdefect$(Q, t, k + 1)$) *if and only if* leftgatedifference$(Q, t, k) =$ true.

*Proof.* The goal follows from gatedifferenceiffflip$(Q, t, k)$. □

**Theorem 2.19.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leftdefect$(Q, t, k+1) =$ true*, then* leftdefect$(Q, t+1, k) =$ false *if and only if* leftgatedifference$(Q, t, k) =$ true*.*

*Proof.* The goal follows from incomingdefecterasediff$(Q, t, k)$. □

**Theorem 2.20.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leftdefect$(Q, t, k+1) =$ true*, then if* leftdefect$(Q, t+1, k) =$ false*, then* collisioncode$(Q, t, k) = 0$*.*

*Proof.* The goal follows from erasurehasnocollision$(Q, t, k)$. □

**Theorem 2.21.** *For every natural number $Q$ and natural number $k$, if $0 < k$, then* leftcolour$(Q, 0, k) = 0$ *or* leftcolour$(Q, 0, k) = 2$*.*

*Proof.* The goal follows from seedleftcolours$(Q, k)$. □

**Theorem 2.22.** *For every natural number $Q$ and natural number $k$, if $2 \leq k$, then* collisioncode$(Q, 0, k) = 0$*.*

*Proof.* The goal follows from seedleftcollisionzero$(Q, k)$. □

**Theorem 2.23.** collisioncode$(1, 1, 2) = 5$*.*

*Proof.* The goal follows from seedadvancingcollisionbirth. □

**Theorem 2.24.** leftdefect$(3, 0, 3) =$ true *and* leftdefect$(3, 1, 2) =$ false $\wedge$ leftgatedifference$(3, 0, 2) =$ true*.*

*Proof.* The goal follows from seedgateerasure. □

**Theorem 2.25.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leftdefect$(Q, t, k) =$ true*, then if* center$(t) =$ true*, then if* center$(t + Q) =$ true*, then* advancingleftcone$(Q, t, k)$*.*

*Proof.* The goal follows from leftdefectblackcone$(Q, t, k)$. □

**Theorem 2.26.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b+r) =$ true*, then there exists a natural number $s$ such that $2 \cdot s \leq r+1$ and $s \leq r \wedge$*advancingleftcollision$(Q, b+s, r+1)$*.*

*Proof.* The goal follows from blackwindowforcesadvancingcollision$(Q, b, r)$. □

**Theorem 2.27.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge$* advancingleftcollision$(m \cdot P, t, P)$.

*Proof.* The goal follows from periodicadvancingleftcollisionwindow$(T, P, m, N)$. □

**Theorem 2.28.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentadvancingleftcollision$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentadvancingleftcolumn$(T, P, m)$. □

**Theorem 2.29.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftdefect$(Q, t, k) =$ false *if and only if* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 3$.

*Proof.* The goal follows from leftdefectfalsecolours$(Q, t, k)$. □

**Theorem 2.30.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftdefect$(Q, t, k) =$ true *if and only if* leftcolour$(Q, t, k) = 1$ *or* leftcolour$(Q, t, k) = 2$.

*Proof.* The goal follows from leftdefecttruecolours$(Q, t, k)$. □

**Theorem 2.31.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $t < k$, then* leftcolour$(Q, t, k) = 0$ *or* leftcolour$(Q, t, k) = 2$.

*Proof.* The goal follows from leftcoloursoutsideseedcone$(Q, t, k)$. □

**Theorem 2.32.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then $k \leq t + 1$.*

*Proof.* The goal follows from collisioncodesupport$(Q, t, k)$. □

**Theorem 2.33.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* gatecode$(Q, t, k) \neq 0$ *if and only if* leftgatedifference$(Q, t, k) =$ true.

*Proof.* The goal follows from gatecodenonzero$(Q, t, k)$. □

**Theorem 2.34.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* $\mathrm{gatecode}(Q, t, k) \leq 6$.

*Proof.* The goal follows from $\mathrm{gatecodebound}(Q, t, k)$. □

**Theorem 2.35.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then* $\mathrm{gatecode}(Q, t, k) = 1 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0$ *and* $(\mathrm{gatecode}(Q, t, k) = 2 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 3 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 1 \wedge \mathrm{leftcolour}(Q, t, k-1) = 1) \wedge (\mathrm{gatecode}(Q, t, k) = 4 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 0) \wedge (\mathrm{gatecode}(Q, t, k) = 5 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 0 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2) \wedge (\mathrm{gatecode}(Q, t, k) = 6 \leftrightarrow \mathrm{leftcolour}(Q, t, k) = 2 \wedge \mathrm{leftcolour}(Q, t, k-1) = 2)$.

*Proof.* The goal follows from $\mathrm{gatecodecolours}(Q, t, k)$. □

**Theorem 2.36.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leftdefect}(Q, t, k+1) = \mathrm{true}$*, then if* $\mathrm{leftdefect}(Q, t+1, k) = \mathrm{false}$*, then* $\mathrm{erasureevent}(Q, t, k)$.

*Proof.* The goal follows from $\mathrm{erasureeventofstep}(Q, t, k)$. □

**Theorem 2.37.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 3$ *if and only if* $\mathrm{gatecode}(Q, t, k) \leq 3$ *and* $\mathrm{leftcolour}(Q, t, k+1) = 2$ *or* $4 \leq \mathrm{gatecode}(Q, t, k)$ *and* $\mathrm{leftcolour}(Q, t, k+1) = 1$.

*Proof.* The goal follows from $\mathrm{erasureblackphaseiff}(Q, t, k)$. □

**Theorem 2.38.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $t+1 < k$, then if* $\mathrm{erasureevent}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+1, k) = 0$.

*Proof.* The goal follows from $\mathrm{erasureoutsideseedconewhite}(Q, t, k)$. □

**Theorem 2.39.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $2 \leq k$, then if* $\mathrm{advancingatdepth}(Q, t, k)$*, then if* $\mathrm{leftdefect}(Q, t+k, 0) = \mathrm{false}$*, then there exists a natural number $j$ such that* $\mathrm{firstpatherasure}(Q, t, k, j)$.

*Proof.* The goal follows from $\mathrm{advancingsourcefirsterasure}(Q, t, k)$. □

**Theorem 2.40.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then if $T \leq N$, then there exists a natural number $u$ such that there exists a natural number $d$ such that $N \leq u$ and $u + 1 \leq 2 \cdot N + m \cdot P + 3 \cdot P \wedge d + 2 \leq P \wedge$* $\mathrm{erasureevent}(m \cdot P, u, d)$.

*Proof.* The goal follows from periodicerasurewindow$(T, P, m, N)$. □

**Theorem 2.41.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$, *then* recurrentsourceerasure$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentsourceerasure$(T, P, m)$. □

**Theorem 2.42.** noperiod *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from noperiodiffnorecurrentcentralcollision. □

**Theorem 2.43.** norowcenterperiod *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from norowcenterperiodiffnorecurrentcentralcollision. □

**Theorem 2.44.** problemone *if and only if* norecurrentcentralcollision.

*Proof.* The goal follows from problemoneiffnorecurrentcentralcollision. □

**Theorem 2.45.** *If* norecurrentcentralcollision, *then* noperiod.

*Proof.* Assume $h$. The goal follows from noperiodofnorecurrentcentralcollision$(h)$. □

**Theorem 2.46.** *For every natural number $T$ and natural number $P$, if $0 < P$, then if* norecurrentcentralcollision, *then not* centerperiodic$(T, P)$.

*Proof.* The goal follows from notcenterperiodicofnorecurrentcentralcollision$(T, P)$. □

**Theorem 2.47.** *If* norecurrentcentralcollision, *then* problemone.

*Proof.* Assume $h$. The goal follows from noteventuallyperiodicofnorecurrentcentralcollision$(h)$. □

**Theorem 2.48.** *For every natural number $b$ and natural number $d$ and natural number $i$,* pairphase(wittphi$(b)$, wittxor(wittphi$(d)$, wittshl(crosscarry$(b, d), 1)), i) =$ bitvalue(rowbit(wittphi(wittxor$(b, d)), i)) -$ bitvalue(rowbit(wittphi$(b), i)$).

*Proof.* The goal follows from pairphasestep$(b, d, i)$. □

**Theorem 2.49.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* alignedphase$(Q, t, t + k + Q) =$ leftphase$(Q, t, k)$.

*Proof.* The goal follows from alignedphaseleft$(Q, t, k)$. □

**Theorem 2.50.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t, k) = 0$ *if and only if* leftdefect$(Q, t, k) =$ false*.*

*Proof.* The goal follows from phasezeroiff$(Q, t, k)$. □

**Theorem 2.51.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t, k) = -1$ *or* leftphase$(Q, t, k) = 1$ *if and only if* leftdefect$(Q, t, k) =$ true*.*

*Proof.* The goal follows from phaseunitiff$(Q, t, k)$. □

**Theorem 2.52.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t+1, k) =$ leftphase$(Q, t, k+1)+$leftphaseflux$(Q, t, k)$.

*Proof.* The goal follows from leftphasebalance$(Q, t, k)$. □

**Theorem 2.53.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* collisioncode$(Q, t, k) \neq 0$*, then* leftphaseflux$(Q, t, k) = -2 \cdot$ leftphase$(Q, t, k + 1)$*.*

*Proof.* The goal follows from collisionphaseflip$(Q, t, k)$. □

**Theorem 2.54.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < k$, then if* advancingatdepth$(Q, t, k)$*, then* leftphase$(Q, t+1, k-1) = -$ leftphase$(Q, t, k)$*.*

*Proof.* The goal follows from advancingphaseflip$(Q, t, k)$. □

**Theorem 2.55.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* gatecode$(Q, t, k) = 1 \to$ leftgatephase$(Q, t, k) = -1$ *and* (gatecode$(Q, t, k) = 2 \to$ leftgatephase$(Q, t, k) = -1) \wedge$ (gatecode$(Q, t, k) = 3 \to$ leftgatephase$(Q, t, k) = -1) \wedge$(gatecode$(Q, t, k) = 4 \to$ leftgatephase$(Q, t, k) = 1) \wedge$(gatecode$(Q, t, k) = 5 \to$ leftgatephase$(Q, t, k) = 1) \wedge$(gatecode$(Q, t, k) = 6 \to$ leftgatephase$(Q, t, k) = 1)$.

*Proof.* The goal follows from gatephasecases$(Q, t, k)$. □

**Theorem 2.56.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* erasureevent$(Q, t, k)$*, then* leftphase$(Q, t + 1, k) = 0$ *and* leftphaseflux$(Q, t, k) = -$ leftphase$(Q, t, k+1) \wedge$ (leftphase$(Q, t, k+1) = -1 \vee$ leftphase$(Q, t, k + 1) = 1)$*.*

*Proof.* The goal follows from erasurephasebalance$(Q, t, k)$. □

**Theorem 2.57.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* erasureevent$(Q, t, k)$*, then* leftcolour$(Q, t+1, k) = 3 \leftrightarrow$ leftgatephase$(Q, t, k) \cdot$leftphase$(Q, t, k+1) = -1$ *and* leftcolour$(Q, t+1, k) = 0 \leftrightarrow$ leftgatephase$(Q, t, k) \cdot$ leftphase$(Q, t, k+1) = 1$.

*Proof.* The goal follows from erasureresetproduct$(Q, t, k)$. □

**Theorem 2.58.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $n$,* pathphase$(Q, t, k, n) =$ leftphase$(Q, t, k)+$ pathflux$(Q, t, k, n)$.

*Proof.* The goal follows from pathphasebalance$(Q, t, k, n)$. □

**Theorem 2.59.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* leftphase$(Q, t, t + Q) = 1$.

*Proof.* The goal follows from seedleftphase$(Q, t)$. □

**Theorem 2.60.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* leftphase$(Q, t + (t + Q), 0) = 1 +$ pathflux$(Q, t, t + Q, t + Q)$.

*Proof.* The goal follows from seedcenterphasebalance$(Q, t)$. □

**Theorem 2.61.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then* pathflux$(m \cdot P, N, N + m \cdot P, N + m \cdot P) = -1$.

*Proof.* The goal follows from periodicseedflux$(T, P, m, N)$. □

**Theorem 2.62.** leftphase$(3, 0, 3) = 1$ *and* leftphase$(3, 1, 2) = 0 \wedge$leftphaseflux$(3, 0, 2) = -1 \wedge$ leftgatephase$(3, 0, 2) = 1$.

*Proof.* The goal follows from seedphasecancellation. □

**Theorem 2.63.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $s$, if* rightmostleft$(Q, t, k)$*, then if $s < k$, then if* centercleanthrough$(Q, t, s)$*, then* rightmostleft$(Q, t + s, k - s)$.

*Proof.* The goal follows from rightmostleftprofile$(Q, t, k, s)$. □

**Theorem 2.64.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leadingadvance$(Q, t, k)$*, then if* centercleanthrough$(Q, t, k)$*, then* firstpatherasure$(Q, t, k, k - 2)$ *and* erasureevent$(Q, t + (k - 1), 0) \wedge (\text{gatecode}(Q, t + (k-1), 0) = 2 \vee \text{gatecode}(Q, t + (k-1), 0) = 5) \wedge \text{center}(t + (k - 1)) = \text{false}$.

*Proof.* The goal follows from leadingfirsterasure$(Q, t, k)$. □

**Theorem 2.65.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* leftadjacentdefect$(Q, b + r) =$ true*, then there exists a natural number $s$ such that there exists a natural number $k$ such that $2 \cdot s \leq r + 1$ and $s \leq r \wedge k + s \leq r + 1 \wedge$* leadingadvance$(Q, b + s, k)$.

*Proof.* The goal follows from blackwindowforcesleadingsource$(Q, b, r)$. □

**Theorem 2.66.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that there exists a natural number $k$ such that $N \leq t$ and $t + 1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k)$.

*Proof.* The goal follows from periodicleadingsourcewindow$(T, P, m, N)$. □

**Theorem 2.67.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $T \leq N$, then there exists a natural number $t$ such that there exists a natural number $k$ such that $N \leq t$ and $t+1 \leq 2 \cdot N + m \cdot P + 2 \cdot P \wedge k \leq P \wedge$* leadingadvance$(m \cdot P, t, k) \wedge$ firstpatherasure$(m \cdot P, t, k, k - 2) \wedge (\text{gatecode}(m \cdot P, t + (k-1), 0) = 2 \vee \text{gatecode}(m \cdot P, t + (k-1), 0) = 5)$.

*Proof.* The goal follows from periodicleadingerasurewindow$(T, P, m, N)$. □

**Theorem 2.68.** *For every natural number $T$ and natural number $P$ and natural number $m$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then* recurrentleadingerasure$(m \cdot P, P)$.

*Proof.* The goal follows from periodicrecurrentleadingerasure$(T, P, m)$. □

**Theorem 2.69.** *For every natural number $Q$ and natural number $R$ and natural number $t$ and natural number $k$,* leftphase$(Q+R, t, k) =$ leftphase$(Q, t, k)+$ leftphase$(R, t + Q, k)$.

*Proof.* The goal follows from leftphaselagadd$(Q, R, t, k)$. □

**Theorem 2.70.** *For every natural number $Q$ and natural number $R$ and natural number $t$ and natural number $k$ and natural number $n$,* $\mathrm{pathflux}(Q + R, t, k, n) = \mathrm{pathflux}(Q, t, k, n) + \mathrm{pathflux}(R, t + Q, k, n)$.

*Proof.* The goal follows from $\mathrm{pathfluxlagadd}(Q, R, t, k, n)$. □

**Theorem 2.71.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leadingadvance}(Q, t, k)$*, then if* $\mathrm{centercleanthrough}(Q, t, k)$*, then* $\mathrm{leftcolour}(Q, t+k, 0) = 3 \leftrightarrow \mathrm{leftgatephase}(Q, t+(k-1), 0) \cdot (\mathrm{leftphase}(Q, t, k) + \mathrm{pathflux}(Q, t, k, k-1)) = -1$ *and* $\mathrm{leftcolour}(Q, t+k, 0) = 0 \leftrightarrow \mathrm{leftgatephase}(Q, t + (k-1), 0) \cdot (\mathrm{leftphase}(Q, t, k) + \mathrm{pathflux}(Q, t, k, k-1)) = 1$.

*Proof.* The goal follows from $\mathrm{leadingphasereset}(Q, t, k)$. □

**Theorem 2.72.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{leftdefect}(Q, t + 1, 0) = \mathrm{false}$*, then* $\mathrm{erasureevent}(Q, t, 0)$ *if and only if* $\mathrm{gatecode}(Q, t, 0) = 2$ *or* $\mathrm{gatecode}(Q, t, 0) = 5$.

*Proof.* The goal follows from $\mathrm{centererasureiffgate}(Q, t)$. □

**Theorem 2.73.** *For every natural number $Q$ and natural number $b$ and natural number $n$ and natural number $i$ and natural number $j$ and natural number $k$, if* $\mathrm{commonwhiteblock}(Q, b, n)$*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* $\mathrm{gatecode}(Q, b + i, 0) = 2$ *or* $\mathrm{gatecode}(Q, b + i, 0) = 5$*, then if* $\mathrm{gatecode}(Q, b + k, 0) = 2$ *or* $\mathrm{gatecode}(Q, b + k, 0) = 5$*, then* $\mathrm{gatecode}(Q, b + j, 0) = \mathrm{gatecode}(Q, b + i, 0)$.

*Proof.* The goal follows from $\mathrm{whiteblockgateconvex}(Q, b, n, i, j, k)$. □

**Theorem 2.74.** *For every natural number $Q$ and natural number $b$ and natural number $n$ and natural number $i$ and natural number $j$ and natural number $k$, if* $\mathrm{commonwhiteblock}(Q, b, n)$*, then if* $\mathrm{centercleanthrough}(Q, b, n+1)$*, then if $i \leq j$, then if $j \leq k$, then if $k \leq n$, then if* $\mathrm{erasureevent}(Q, b + i, 0)$*, then if* $\mathrm{erasureevent}(Q, b + k, 0)$*, then* $\mathrm{erasureevent}(Q, b + j, 0)$ *and* $\mathrm{gatecode}(Q, b + j, 0) = \mathrm{gatecode}(Q, b + i, 0)$.

*Proof.* The goal follows from $\mathrm{whiteblockerasureconvex}(Q, b, n, i, j, k)$. □

**Theorem 2.75.** $\mathrm{leadingadvance}(4, 5, 2)$ *and* $\mathrm{centercleanthrough}(4, 5, 2) \wedge \mathrm{gatecode}(4, 6, 0) = 5$.

*Proof.* The goal follows from seedleadinggatefive. □

**Theorem 2.76.** $\mathrm{leadingadvance}(7, 7, 4)$ *and* $\mathrm{centercleanthrough}(7, 7, 4) \wedge \mathrm{gatecode}(7, 10, 0) = 2$.

*Proof.* The goal follows from seedleadinggatetwo. □

**Theorem 2.77.** leadingadvance$(11, 4, 4)$ *and* centercleanthrough$(11, 4, 4) \wedge$ gatecode$(11, 7, 0) = 5 \wedge$ leftcolour$(11, 8, 0) = 3$.

*Proof.* The goal follows from seedleadingblackreset. □

**Theorem 2.78.** *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* recoverphase$(incomingBit, laterBit,$ transferphase$(incomingBi$ *incoming.*

*Proof.* The goal follows from transferrecover$(incomingBit, laterBit, incoming, gate)$. □

**Theorem 2.79.** *For every truth value incomingBit and truth value laterBit and integer incoming and integer gate,* transferphase$(incomingBit, laterBit, incoming, gate) =$ $0$ *if and only if incoming* $=$ recoverphase$(incomingBit, laterBit, 0, gate)$.

*Proof.* The goal follows from transferkernel$(incomingBit, laterBit, incoming, gate)$. □

**Theorem 2.80.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t{+}1, k) =$ transferphase(leftcell$(t, k{+}1)$, latergate$(Q, t, k)$, leftphase$(Q, t, k{+}1)$, leftgatephase$(Q, t, k)$).

*Proof.* The goal follows from phasetransfer$(Q, t, k)$. □

**Theorem 2.81.** *For every natural number $Q$ and natural number $t$ and natural number $k$,* leftphase$(Q, t, k{+}1) =$ recoverphase(leftcell$(t, k{+}1)$, latergate$(Q, t, k)$, leftphase$(Q, t{+}1, k)$, leftgatephase$(Q, t, k)$).

*Proof.* The goal follows from phaserecover$(Q, t, k)$. □

**Theorem 2.82.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* erasureevent$(Q, t, k)$*, then not* leftgatephase$(Q, t, k) = 0$ *and* leftphase$(Q, t, k{+}1) =$ recoverphase(leftcell$(t, k{+}1)$, latergate$(Q, t, k), 0,$ leftgatephase$(Q, t, k)$).

*Proof.* The goal follows from erasurerecovery$(Q, t, k)$. □

**Theorem 2.83.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $g = 2$ or $g = 5$, then there exists a natural number $t$ such that $N \leq t$ and* erasureevent$(m \cdot P, t, 0) \wedge$ gatecode$(m \cdot P, t, 0) = g$.

*Proof.* The goal follows from periodicerasureorientation$(T, P, m, N, g)$. □

**Theorem 2.84.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* leadingadvance$(Q, t, k)$*, then if* centercleanthrough$(Q, t, k)$*, then there exists a natural number $s$ such that $s + 1 < k$ and* center$(t + s) =$ true*.*

*Proof.* The goal follows from leadingblackbeforearrival$(Q, t, k)$. □

**Theorem 2.85.** *For every natural number $Q$ and natural number $b$ and natural number $r$, if $0 < r$, then if* centerwindowagree$(b, b + Q, r + 1)$*, then if* center$(b) =$ true*, then if* whitecentergap$(b, r)$*, then if* leftadjacentdefect$(Q, b+ r) =$ true*, then there exists a natural number $k$ such that $k \leq r + 1$ and* leadingadvance$(Q, b, k)$*.*

*Proof.* The goal follows from blackgapforcesinitialleading$(Q, b, r)$. □

**Theorem 2.86.** *For every natural number $Q$ and natural number $t$, if* leadingadvance$(Q, t, 2)$*, then if* centercleanthrough$(Q, t, 2)$*, then* center$(t) =$ true*.*

*Proof.* The goal follows from leadingdepthtwoblack$(Q, t)$. □

**Theorem 2.87.** *For every natural number $Q$ and natural number $b$ and natural number $r$ and natural number $g$, if $0 < r$, then if* centerwindowagree$(b, b+ Q, r + 1)$*, then if* center$(b) =$ true*, then if* whitecentergap$(b, r)$*, then if* erasureevent$(Q, b + r, 0)$*, then if* gatecode$(Q, b + r, 0) = g$*, then there exists a natural number $k$ such that $k \leq r + 1$ and* leadingadvance$(Q, b, k) \wedge$ firstpatherasure$(Q, b, k, k - 2) \wedge$ gatecode$(Q, b + (k - 1), 0) = g$*.*

*Proof.* The goal follows from blackgaperasureorigin$(Q, b, r, g)$. □

**Theorem 2.88.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $g = 2$ or $g = 5$, then there exists a natural number $b$ such that there exists a natural number $k$ such that $N \leq b$ and $k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k - 1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k-2) \wedge$ gatecode$(m \cdot P, b + (k - 1), 0) = g$*.*

*Proof.* The goal follows from periodicblackerasureorigin$(T, P, m, N, g)$. □

**Theorem 2.89.** *For every natural number $t$,* rightcell$(t+1, 1) =$ bxor(center$(t)$, bxor(rightcell$(t, 1)$, right

*Proof.* The goal follows from rightimportstep$(t)$. □

**Theorem 2.90.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{center}(t+Q) = \mathrm{center}(t)$*, then* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{bxor}(\mathrm{rightdefect}(Q, t, 1), \mathrm{rightimportdifference}(Q, t))$.

*Proof.* The goal follows from $\mathrm{rightdefectimportbalance}(Q, t)$. □

**Theorem 2.91.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{center}(t) = \mathrm{true}$*, then if* $\mathrm{center}(t+Q) = \mathrm{true}$*, then if* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{true}$*, then if* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{true}$*, then* $\mathrm{rightphase}(Q, t+1, 1) = -\mathrm{rightphase}(Q, t, 1)$.

*Proof.* The goal follows from $\mathrm{blackresetsurvival}(Q, t)$. □

**Theorem 2.92.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{center}(t) = \mathrm{true}$*, then if* $\mathrm{center}(t+Q) = \mathrm{true}$*, then if* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{false}$*, then if* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{true}$*, then* $\mathrm{rightcell}(t, 1) = \mathrm{false}$ *and* $\mathrm{rightcell}(t+Q, 1) = \mathrm{false} \wedge \mathrm{rightdefect}(Q, t, 2) = \mathrm{true} \wedge \mathrm{rightphase}(Q, t+1, 1) = -\mathrm{rightphase}(Q, t, 2)$.

*Proof.* The goal follows from $\mathrm{blackresetimport}(Q, t)$. □

**Theorem 2.93.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and truth value $before$, if $0 < P$, then if $0 < m$, then if* $\mathrm{centerperiodic}(T, P)$*, then there exists a natural number $t$ such that $N \leq t$ and* $\mathrm{rightimport}(t) = before \wedge \mathrm{rightimport}(t + m \cdot P) = \mathrm{not}(before) \wedge \mathrm{rightdefect}(m \cdot P, t+1, 1) = \mathrm{not}(\mathrm{rightdefect}(m \cdot P, t, 1))$.

*Proof.* The goal follows from $\mathrm{periodicrightimportorientation}(T, P, m, N, before)$. □

**Theorem 2.94.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{center}(t+Q) = \mathrm{center}(t)$*, then if* $\mathrm{rightdefect}(Q, t, 1) = \mathrm{false}$*, then if* $\mathrm{rightdefect}(Q, t+1, 1) = \mathrm{true}$*, then* $\mathrm{rightcell}(t, 1) = \mathrm{false}$ *and* $\mathrm{rightcell}(t+Q, 1) = \mathrm{false} \wedge \mathrm{rightdefect}(Q, t, 2) = \mathrm{true} \wedge \mathrm{rightimportdifference}(Q, t) = \mathrm{true} \wedge \mathrm{rightphase}(Q, t+1, 1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(t))) \cdot \mathrm{rightphase}(Q, t, 2)$.

*Proof.* The goal follows from $\mathrm{rightheadbirth}(Q, t)$. □

**Theorem 2.95.** *For every natural number $Q$ and natural number $start$ and natural number $finish$, if* $\mathrm{headimportpath}(Q, start, finish)$*, then if* $\mathrm{centercleanthrough}(Q, start, finish - start)$*, then* $\mathrm{rightphase}(Q, finish, 1) = \mathrm{centersign}(start, finish - start) \cdot \mathrm{rightphase}(Q, start, 2)$.

*Proof.* The goal follows from $\mathrm{headimportpathphase}(Q, start, finish)$. □

**Theorem 2.96.** *For every natural number $Q$ and natural number $t$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{leftdefect}(Q, t+1, 0) = \mathrm{false}$*, then if* $\mathrm{center}(t) = \mathrm{false}$*, then* $\mathrm{leftphase}(Q, t, 1) = (1 - 2 \cdot \mathrm{bitvalue}(\mathrm{center}(t+1))) \cdot \mathrm{rightphase}(Q, t, 1)$.

*Proof.* The goal follows from whitecenterphasematching$(Q, t)$. □

**Theorem 2.97.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $g = 2$ or $g = 5$, then there exists a natural number start such that there exists a natural number $b$ such that there exists a natural number $k$ such that $N \leq start$ and $N \leq b \wedge k \leq P \wedge$* center$(b) =$ true $\wedge$ whitecentergap$(b, k-1) \wedge$ leadingadvance$(m \cdot P, b, k) \wedge$ firstpatherasure$(m \cdot P, b, k, k-2) \wedge$ gatecode$(m \cdot P, b + (k-1), 0) = g \wedge$headimportpath$(m \cdot P, start, b + (k-1)) \wedge$leftphase$(m \cdot P, b, k) +$ pathflux$(m \cdot P, b, k, k-1) = (1 - 2 \cdot$ bitvalue(center$(b + k))) \cdot$ centersign$(start, b + (k - 1) - start) \cdot$ rightphase$(m \cdot P, start, 2)$.

*Proof.* The goal follows from periodiccoupledcancellation$(T, P, m, N, g)$. □

**Theorem 2.98.** *For every natural number $Q$ and natural number $b$ and natural number $k$, if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then* leftphase$(Q, b, k) +$ pathflux$(Q, b, k, k-1) = (1 - 2 \cdot$ bitvalue(center$(b + k))) \cdot$ leftgatephase$(Q, b + (k-1), 0)$.

*Proof.* The goal follows from leadingarrivalphaseeqgatephase$(Q, b, k)$. □

**Theorem 2.99.** *For every natural number $Q$ and natural number start and natural number $finish$, if* headimportpath$(Q, start, finish)$*, then if* centercleanthrough$(Q, start, finish - start)$*, then if* center$(finish) =$ false*, then* leftgatephase$(Q, finish, 0) =$ centersign$(start, finish - start) \cdot$rightphase$(Q, start, 2)$.

*Proof.* The goal follows from headimportpathgatephase$(Q, start, finish)$. □

**Theorem 2.100.** *For every natural number $Q$ and natural number start and natural number $finish$, if* headimportpath$(Q, start, finish)$*, then if* centercleanthrough$(Q, start, finish - start)$*, then if* center$(finish) =$ false*, then not* leftgatephase$(Q, finish, 0) = -($centersign$(start, finish - start) \cdot$ rightphase$(Q, start, 2))$.

*Proof.* The goal follows from headimportpathrejectsnegativegate$(Q, start, finish)$. □

**Theorem 2.101.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if $0 < P$, then if $0 < m$, then if* centerperiodic$(T, P)$*, then if $g = 2$ or $g = 5$, then there exists a natural number start such that there exists a natural number*

$b$ *such that there exists a natural number* $k$ *such that* $N \leq$ *start and* $N \leq b \wedge k \leq P \wedge \mathrm{center}(b) = \mathrm{true} \wedge \mathrm{whitecentergap}(b, k-1) \wedge \mathrm{leadingadvance}(m \cdot P, b, k) \wedge \mathrm{firstpatherasure}(m \cdot P, b, k, k-2) \wedge \mathrm{gatecode}(m \cdot P, b+(k-1), 0) = g \wedge \mathrm{headimportpath}(m \cdot P, \mathit{start}, b+(k-1)) \wedge \mathrm{leftgatephase}(m \cdot P, b+(k-1), 0) = \mathrm{centersign}(\mathit{start}, b+(k-1)-\mathit{start}) \cdot \mathrm{rightphase}(m \cdot P, \mathit{start}, 2)$.

*Proof.* The goal follows from $\mathrm{periodiccoupledgatephasesign}(T, P, m, N, g)$. □

**Theorem 2.102.** *For every natural number* $T$ *and natural number* $P$ *and natural number* $m$ *and natural number* $N$*, if* $0 < P$*, then if* $0 < m$*, then if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq N$*, then* $\mathrm{pathflux}(m \cdot P + m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$ *and* $\mathrm{pathflux}(m \cdot P, N, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = 0 \wedge \mathrm{pathflux}(m \cdot P, N + m \cdot P, N + (m \cdot P + m \cdot P), N + (m \cdot P + m \cdot P)) = -1$.

*Proof.* The goal follows from $\mathrm{periodicdoubledseedfluxsplit}(T, P, m, N)$. □

**Theorem 2.103.** $\mathrm{leadingadvance}(11, 4, 4)$ *and* $\mathrm{centercleanthrough}(11, 4, 4) \wedge \mathrm{headimportpath}(11, 6, 7) \wedge \mathrm{centercleanthrough}(11, 6, 1) \wedge \mathrm{leftgatephase}(11, 7, 0) = 1 \wedge \mathrm{centersign}(6, 1) = 1 \wedge \mathrm{rightphase}(11, 6, 2) = 1$.

*Proof.* The goal follows from seedcoupledgatesign. □

**Theorem 2.104.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $j$*, if* $\mathrm{rightdefect}(Q, t+1, j+1) = \mathrm{true}$*, then* $\mathrm{rightdefect}(Q, t, j) = \mathrm{true}$ *or* $\mathrm{rightdefect}(Q, t, j+1) = \mathrm{true} \vee \mathrm{rightdefect}(Q, t, j+2) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{rightdefecthasancestor}(Q, t, j)$. □

**Theorem 2.105.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $j$ *and natural number* $T$*, if* $1 \leq j$*, then if* $T \leq t$*, then if* $\mathrm{rightdefect}(Q, t, j) = \mathrm{true}$*, then if* $\mathrm{centercleanthrough}(Q, T, t-T)$*, then* $\mathrm{rightlineage}(Q, t, j, T)$.

*Proof.* The goal follows from $\mathrm{rightlineageofcleancenter}(Q, t, j, T)$. □

**Theorem 2.106.** *For every natural number* $Q$ *and natural number* $t$ *and natural number* $k$ *and natural number* $T$*, if* $1 \leq k$*, then if* $T \leq t$*, then if* $\mathrm{leftdefect}(Q, t, k) = \mathrm{true}$*, then if* $\mathrm{centercleanthrough}(Q, T, t-T)$*, then* $\mathrm{leftlineage}(Q, t, k, T)$.

*Proof.* The goal follows from $\mathrm{leftlineageofcleancenter}(Q, t, k, T)$. □

**Theorem 2.107.** *For every natural number $Q$ and natural number $t$ and natural number $j$ and natural number $T$, if $T \leq t$, then if* $\mathrm{rightlineage}(Q, t, j, T)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq j + (t - T) \wedge j \leq i + (t - T) \wedge \mathrm{rightdefect}(Q, T, i) = \mathrm{true}$.*

*Proof.* The goal follows from $\mathrm{rightlineageatstart}(Q, t, j, T)$. □

**Theorem 2.108.** *For every natural number $Q$ and natural number $t$ and natural number $k$ and natural number $T$, if $T \leq t$, then if* $\mathrm{leftlineage}(Q, t, k, T)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq k + (t - T) \wedge k \leq i + (t - T) \wedge \mathrm{leftdefect}(Q, T, i) = \mathrm{true}$.*

*Proof.* The goal follows from $\mathrm{leftlineageatstart}(Q, t, k, T)$. □

**Theorem 2.109.** *For every natural number $Q$ and natural number $t$ and natural number $j$, if* $\mathrm{rightlineage}(Q, t, j, 0)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq Q \wedge i \leq j + t \wedge j \leq i + t \wedge \mathrm{rightdefect}(Q, 0, i) = \mathrm{true} \wedge \mathrm{rightcell}(Q, i) = \mathrm{true} \wedge \mathrm{rightphase}(Q, 0, i) = 1$.*

*Proof.* The goal follows from $\mathrm{rightlineageseed}(Q, t, j)$. □

**Theorem 2.110.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if* $\mathrm{leftlineage}(Q, t, k, 0)$*, then there exists a natural number $i$ such that $1 \leq i$ and $i \leq Q \wedge i \leq k + t \wedge k \leq i + t \wedge \mathrm{leftdefect}(Q, 0, i) = \mathrm{true} \wedge \mathrm{leftcell}(Q, i) = \mathrm{true} \wedge \mathrm{leftphase}(Q, 0, i) = 1$.*

*Proof.* The goal follows from $\mathrm{leftlineageseed}(Q, t, k)$. □

**Theorem 2.111.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then there exists a natural number $y$ such that $y \leq t + Q$ and* $\mathrm{rightfront}(Q, t, y)$*.*

*Proof.* The goal follows from $\mathrm{existsrightfront}(Q, t)$. □

**Theorem 2.112.** *For every natural number $Q$ and natural number $t$ and natural number $y$ and natural number $z$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if* $\mathrm{rightfront}(Q, t + 1, z)$*, then $y \leq z + 1$.*

*Proof.* The goal follows from $\mathrm{rightfrontlower}(Q, t, y, z)$. □

**Theorem 2.113.** *For every natural number $Q$ and natural number $t$ and natural number $y$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if $2 \leq y$, then* $\mathrm{rightdefect}(Q, t + 1, y - 1) = \mathrm{not}(\mathrm{rightcell}(t, y - 1))$*.*

*Proof.* The goal follows from $\mathrm{rightfrontdescent}(Q, t, y)$. □

**Theorem 2.114.** *For every natural number $Q$ and natural number $t$ and natural number $y$ and natural number $z$, if* $\mathrm{leftdefect}(Q, t, 0) = \mathrm{false}$*, then if* $\mathrm{rightfront}(Q, t, y)$*, then if* $2 \leq y$*, then if* $\mathrm{rightfront}(Q, t + 1, z)$*, then if* $z + 1 \leq y$*, then* $z = y - 1$ *and* $\mathrm{rightcell}(t, y - 1) = \mathrm{false}$*.*

*Proof.* The goal follows from $\mathrm{rightfrontlockstep}(Q, t, y, z)$. □

**Theorem 2.115.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number $s$, if* $\mathrm{rightmostleft}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $s+1 < k$*, then if* $\mathrm{rightdefect}(Q, b+s, 1) = \mathrm{true}$*, then* $\mathrm{center}(b + s) = \mathrm{true}$*.*

*Proof.* The goal follows from $\mathrm{leadingheadblackbeforearrival}(Q, b, k, s)$. □

**Theorem 2.116.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number $start$, if* $3 \leq k$*, then if* $\mathrm{leadingadvance}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $\mathrm{whitecentergap}(b, k - 1)$*, then if* $\mathrm{headimportpath}(Q, start, b + (k - 1))$*, then* $start = b + (k - 2)$*.*

*Proof.* The goal follows from $\mathrm{coupledimportbornatarrival}(Q, b, k, start)$. □

**Theorem 2.117.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number $s$, if* $0 < Q$*, then if* $3 \leq k$*, then if* $\mathrm{leadingadvance}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $\mathrm{whitecentergap}(b, k-1)$*, then if* $1 \leq s$*, then if* $s+1 < k$*, then there exists a natural number $y$ such that* $\mathrm{rightdefect}(Q, b + s, 1) = \mathrm{false}$ *and* $\mathrm{rightfront}(Q, b + s, y) \wedge 2 \leq y \wedge y \leq k - s$*.*

*Proof.* The goal follows from $\mathrm{coupledrightfrontmirrorat}(Q, b, k, s)$. □

**Theorem 2.118.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number $s$, if* $0 < Q$*, then if* $3 \leq k$*, then if* $\mathrm{leadingadvance}(Q, b, k)$*, then if* $\mathrm{centercleanthrough}(Q, b, k)$*, then if* $\mathrm{whitecentergap}(b, k-1)$*, then if* $1 \leq s$*, then if* $s + 2 < k$*, then if* $\mathrm{rightfront}(Q, b + s, k - s)$*, then* $\mathrm{rightcell}(b + s, k - s - 1) = \mathrm{false}$ *and* $\mathrm{rightfront}(Q, b + (s + 1), k - (s + 1))$*.*

*Proof.* The goal follows from $\mathrm{coupledwhitestaircaseat}(Q, b, k, s)$. □

**Theorem 2.119.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $N$ and natural number $g$, if* $0 < P$*, then if* $0 < m$*, then if* $\mathrm{centerperiodic}(T, P)$*, then if* $g = 2$ *or* $g = 5$*, then there exists a natural number start such that there exists a natural number $b$ such that there exists a natural number $k$ such that there exists a natural*

*number $y$ such that $N \leq start$ and $N \leq b \wedge T \leq b \wedge T \leq start \wedge 2 \leq k \wedge k \leq P \wedge \mathrm{center}(b) = \mathrm{true} \wedge \mathrm{whitecentergap}(b, k-1) \wedge \mathrm{leadingadvance}(m \cdot P, b, k) \wedge \mathrm{gatecode}(m \cdot P, b + (k-1), 0) = g \wedge \mathrm{headimportpath}(m \cdot P, start, b + (k-1)) \wedge (3 \leq k \to start = b + (k-2)) \wedge (3 \leq k \to \mathrm{rightfront}(m \cdot P, b+1, y) \wedge 2 \leq y \wedge y \leq k-1) \wedge \mathrm{rightlineage}(m \cdot P, start, 2, T) \wedge \mathrm{leftlineage}(m \cdot P, b, k, T)$.*

*Proof.* The goal follows from $\mathrm{periodiccoupledimportancestry}(T, P, m, N, g)$. □

**Theorem 2.120.** $\mathrm{leadingadvance}(14, 5, 3)$ *and* $\mathrm{centercleanthrough}(14, 5, 3) \wedge \mathrm{whitecentergap}(5, 2) \wedge \mathrm{center}(5) = \mathrm{true} \wedge \mathrm{rightfront}(14, 6, 2) \wedge \mathrm{headimportpath}(14, 6, 7)$.

*Proof.* The goal follows from seedcoupledimportatarrival. □

**Theorem 2.121.** *For every natural number $t$ and natural number $k$ and natural number $n$,* $\mathrm{bitvalue}(\mathrm{raybit}(t, k, n)) = \mathrm{bitvalue}(\mathrm{raybit}(t, k, 0)) + \mathrm{rayflux}(t, k, n)$.

*Proof.* The goal follows from $\mathrm{raybitbalance}(t, k, n)$. □

**Theorem 2.122.** *For every natural number $Q$ and natural number $d$ and natural number $t$ and natural number $p$, if $d + p \leq t$, then* $\mathrm{leftphase}(Q + Q, t + (t - d - p), p) = \mathrm{diagonalphase}(Q, d, t) + \mathrm{diagonalflux}(Q, d, t, p)$.

*Proof.* The goal follows from $\mathrm{diagonalraybalance}(Q, d, t, p)$. □

**Theorem 2.123.** *For every natural number $Q$ and natural number $R$ and natural number $d$ and natural number $t$ and natural number $p$,* $\mathrm{diagonalflux}(Q + R, d, t, p) = \mathrm{diagonalflux}(Q, d, t, p) + \mathrm{diagonalflux}(R, d, t + Q, p)$.

*Proof.* The goal follows from $\mathrm{diagonalfluxlagadd}(Q, R, d, t, p)$. □

**Theorem 2.124.** *For every natural number $Q$ and natural number $z$ and natural number $t$, if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{doublesat}(Q, z)$*, then if* $\mathrm{settledby}(z+1, t)$*, then* $\mathrm{diagonalphase}(Q, z+1, t) = -1 \vee \mathrm{diagonalphase}(Q, z+1, t) = 1$ *and* $\mathrm{diagonalphase}(Q, z+1, t+Q) = -\mathrm{diagonalphase}(Q, z+1, t) \wedge \mathrm{diagonalphase}(Q + Q, z+1, t) = 0$.

*Proof.* The goal follows from $\mathrm{diagonalphasedoubles}(Q, z, t)$. □

**Theorem 2.125.** *For every natural number $Q$ and natural number $z$ and natural number $t$, if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{settledby}(z+2, t)$*, then* $\mathrm{diagonalphase}(Q, z+3, t+1) = -\mathrm{diagonalphase}(Q, z+1, t)$.

*Proof.* The goal follows from $\mathrm{diagonalresetphase}(Q, z, t)$. □

**Theorem 2.126.** *For every natural number $Q$ and natural number $z$ and natural number $t$, if* zerodiagonal$(z)$*, then if* settledby$(z+2,t)$*, then if* $z+3 \leq t$*, then* diagonalflux$(Q, z+3, t+1, 1) =$ diagonalflux$(Q, z+1, t, 1) + 2 \cdot$ diagonalphase$(Q, z+1, t)$.

*Proof.* The goal follows from diagonalresetfluxjump$(Q, z, t)$. □

**Theorem 2.127.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $d$ and natural number $t$, if* centerperiodic$(T, P)$*, then if* $d+1 \leq t$*, then if* $T \leq t+(t-d-1)$*, then if* center$(t+(t-d-1)) =$ false*, then* diagonalphase$(m \cdot P, d, t)$ + diagonalflux$(m \cdot P, d, t, 1) = (1 - 2 \cdot$ bitvalue(center$(t+(t-d-1)+1))) \cdot$rightphase$(m \cdot P + m \cdot P, t+(t-d-1), 1)$.

*Proof.* The goal follows from periodicdiagonalwhiteheadbalance$(T, P, m, d, t)$. □

**Theorem 2.128.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $d$ and natural number $t$ and natural number start, if* centerperiodic$(T, P)$*, then if* $d+1 \leq t$*, then if* $T \leq start$*, then if* headimportpath$(m \cdot P + m \cdot P, start, t+(t-d-1))$*, then if* center$(t+(t-d-1)) =$ false*, then* diagonalphase$(m \cdot P, d, t)$ + diagonalflux$(m \cdot P, d, t, 1) = (1 - 2 \cdot$ bitvalue(center$(t + (t-d-1) + 1))) \cdot$ (centersign$(start, t + (t-d-1) - start) \cdot$ rightphase$(m \cdot P + m \cdot P, start, 2))$.

*Proof.* The goal follows from periodicdiagonalimportbalance$(T, P, m, d, t, start)$. □

**Theorem 2.129.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start and natural number $q$, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $1 \leq q$*, then if* $q + 1 < k$*, then* rightfront$(Q, b+1, k-1)$ *and* rightcell$(b+1, q) =$ false $\wedge$ rightcell$(b+1+Q, q) =$ false.

*Proof.* The goal follows from coupledrightfrontexact$(Q, b, k, start, q)$. □

**Theorem 2.130.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start and natural number $i$, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $1 \leq i$*, then if* $i + 1 < k$*, then* rightfront$(Q, b+i, k-i)$ *and* rightcell$(b+i, k-i-1) =$ false $\wedge$ rightcell$(b+i+Q, k-i-1) =$ false.

*Proof.* The goal follows from coupledforcedwhitestaircase$(Q, b, k, start, i)$. □

**Theorem 2.131.** *For every natural number $Q$ and natural number $b$ and natural number start, if* leadingadvance$(Q, b, 2)$*, then if* centercleanthrough$(Q, b, 2)$*, then if* headimportpath$(Q, start, b+1)$*, then* center$(b) =$ true *and* center$(b+1) =$ false $\wedge$($start < b \wedge$rightdefect$(Q, b, 1) =$ true $\wedge$ rightimportdifference$(Q, b) =$ false $\wedge$ leftphase$(Q, b, 2) = (1-2 \cdot$bitvalue(center$(b+2))) \cdot$rightphase$(Q, b, 1) \vee$ $start = b \wedge$ rightdefect$(Q, b, 1) =$ false $\wedge$ rightcell$(b, 1) =$ false $\wedge$ rightcell$(b + Q, 1) =$ false $\wedge$ rightdefect$(Q, b, 2) =$ true $\wedge$ rightimportdifference$(Q, b) =$ true $\wedge$ leftphase$(Q, b, 2) = (1-2 \cdot$bitvalue(center$(b+2))) \cdot$rightphase$(Q, b, 2))$.

*Proof.* The goal follows from depthtwocancellationcases$(Q, b, start)$. □

**Theorem 2.132.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start, if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then* $start = b+(k-2)$ *or* $k = 2$ *and* $start < b \wedge$rightdefect$(Q, b, 1) =$ true.

*Proof.* The goal follows from coupledimportbirthordepthtwosurvival$(Q, b, k, start)$. □

**Theorem 2.133.** *For every natural number $Q$ and natural number $b$ and natural number $k$ and natural number start and natural number $q$, if* $0 < Q$*, then if* $3 \leq k$*, then if* leadingadvance$(Q, b, k)$*, then if* centercleanthrough$(Q, b, k)$*, then if* center$(b) =$ true*, then if* whitecentergap$(b, k-1)$*, then if* headimportpath$(Q, start, b+(k-1))$*, then if* $q+2 < k$*, then* rightcell$(b, q) =$ true *and* rightcell$(b+Q, q) =$ true.

*Proof.* The goal follows from coupledblackrootprefix$(Q, b, k, start, q)$. □

**Theorem 2.134.** leadingadvance$(4, 5, 2)$ *and* centercleanthrough$(4, 5, 2) \wedge$ headimportpath$(4, 2, 6) \wedge$rightdefect$(4, 5, 1) =$ true $\wedge$ rightimportdifference$(4, 5) =$ false.

*Proof.* The goal follows from seeddepthtwosurvival. □

**Theorem 2.135.** *For every natural number $Q$ and natural number $t$,* alignedbackground$(Q, t+1) =$ wittphi(alignedbackground$(Q, t)$).

*Proof.* The goal follows from alignedbackgroundstep$(Q, t)$. □

**Theorem 2.136.** *For every natural number $Q$ and natural number $t$,* $\mathrm{aligneddefect}(Q, t+1) = \mathrm{defectstep}(\mathrm{alignedbackground}(Q, t), \mathrm{aligneddefect}(Q, t))$.

*Proof.* The goal follows from $\mathrm{aligneddefectstep}(Q, t)$. □

**Theorem 2.137.** *For every natural number $Q$ and natural number $t$,* $\mathrm{rowbit}(\mathrm{alignedbackground}(Q, t), 2 \cdot t + Q) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{backgroundtopbit}(Q, t)$. □

**Theorem 2.138.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), 2 \cdot (t + Q)) = \mathrm{true}$.

*Proof.* The goal follows from $\mathrm{aligneddefecttopbit}(Q, t)$. □

**Theorem 2.139.** *For every natural number $Q$ and natural number $t$, if $0 < Q$, then* $\mathrm{twopow}(2 \cdot (t + Q)) \leq \mathrm{aligneddefect}(Q, t)$ *and* $\mathrm{aligneddefect}(Q, t) < \mathrm{twopow}(2 \cdot (t + Q) + 1)$.

*Proof.* The goal follows from $\mathrm{aligneddefectmagnitude}(Q, t)$. □

**Theorem 2.140.** *For every natural number $Q$ and natural number $t$ and natural number $j$, if $2 \cdot (t+Q) < j$, then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), j) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{aligneddefectabovetop}(Q, t, j)$. □

**Theorem 2.141.** *For every natural number $Q$ and natural number $t$ and natural number $j$, if $j < Q$, then* $\mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), j) = \mathrm{rowbit}(\mathrm{wittrow}(t+Q), j)$.

*Proof.* The goal follows from $\mathrm{aligneddefectlowprefix}(Q, t, j)$. □

**Theorem 2.142.** *For every natural number $b$ and natural number $d$,* $\mathrm{defectstep}(b, d) = \mathrm{wittxor}(\mathrm{wittlinear}(d), \mathrm{wittshl}(\mathrm{wittxor}(\mathrm{wittcarry}(d), \mathrm{crosscarry}(b, d)), 1))$.

*Proof.* The goal follows from $\mathrm{defectsteplinear}(b, d)$. □

**Theorem 2.143.** *For every natural number $b$ and natural number $d$,* $\mathrm{rowbit}(\mathrm{defectstep}(b, d), 0) = \mathrm{rowbit}(d, 0)$.

*Proof.* The goal follows from $\mathrm{defectsteplowbit}(b, d)$. □

**Theorem 2.144.** *For every natural number $b$ and natural number $d$,* $\mathrm{vala}(\mathrm{defectstep}(b, d)) = \mathrm{valb}(\mathrm{wittxor}(\mathrm{wittcarry}(d), \mathrm{crosscarry}(b, d)))$ *and* $\mathrm{valb}(\mathrm{defectstep}(b, d)) = \mathrm{bxor}(\mathrm{vala}(\mathrm{wittxor}(\mathrm{wittcarry}(d$

*Proof.* The goal follows from $\mathrm{defectstepvalcoordinates}(b, d)$. □

**Theorem 2.145.** *For every natural number $j$,* $\mathrm{vala}(\mathrm{wittshl}(1, j+3)) = \mathrm{vala}(\mathrm{wittshl}(1, j))$ *and* $\mathrm{valb}(\mathrm{wittshl}(1, j+3)) = \mathrm{valb}(\mathrm{wittshl}(1, j)) \wedge (\mathrm{vala}(\mathrm{wittshl}(1, j)) = \mathrm{true} \vee \mathrm{valb}(\mathrm{wittshl}(1, j)) = \mathrm{true})$.

*Proof.* The goal follows from $\mathrm{edgevalcoordinates}(j)$. □

**Theorem 2.146.** *For every natural number $t$ and natural number $k$,* $\mathrm{aligneddefect}(\mathrm{twopow}(k), t) = \mathrm{wittxor}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), \mathrm{wittxor}(\mathrm{wittrow}(t), \mathrm{wittshl}(\mathrm{wittrow}(t), \mathrm{twopow}(k) + \mathrm{twopow}(k))))$.

*Proof.* The goal follows from $\mathrm{aligneddefectwindow}(t, k)$. □

**Theorem 2.147.** *For every natural number $t$ and natural number $k$, if* $t < \mathrm{twopow}(k)$, *then* $\mathrm{rowbit}(\mathrm{aligneddefect}(\mathrm{twopow}(k), t), 0) = \mathrm{true}$ *and* $\mathrm{rowbit}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), 0) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(\mathrm{twopow}(k), t), 2 \cdot (t + \mathrm{twopow}(k))) = \mathrm{true} \wedge \mathrm{rowbit}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), 2 \cdot (t + \mathrm{twopow}(k))) = \mathrm{false} \wedge \mathrm{vala}(\mathrm{wittwin}(t, \mathrm{twopow}(k))) = \mathrm{vala}(\mathrm{wittrow}(t + \mathrm{twopow}(k))) \wedge \mathrm{valb}(\mathrm{wittwin}(t, \mathrm{twopow}(k))) = \mathrm{valb}(\mathrm{wittrow}(t + \mathrm{twopow}(k))) \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(\mathrm{twopow}(k), t), t + \mathrm{twopow}(k)) = \mathrm{rowbit}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t + \mathrm{twopow}(k))$.

*Proof.* The goal follows from $\mathrm{dyadicwindowreadings}(t, k)$. □

**Theorem 2.148.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T, P)$, *then if* $T \leq t$, *then* $\mathrm{rowbit}(\mathrm{wittphi}(\mathrm{aligneddefect}(m \cdot P, t)), t + m \cdot P + 1) = \mathrm{rowbit}(\mathrm{crosscarry}(\mathrm{alignedbackground}(m \cdot P, t), \mathrm{aligneddefect}(m \cdot P, t)), t + m \cdot P)$.

*Proof.* The goal follows from $\mathrm{periodiccentercrosscarry}(T, P, m, t)$. □

**Theorem 2.149.** $\mathrm{aligneddefect}(1, 0) = 5$ *and* $\mathrm{vala}(\mathrm{aligneddefect}(1, 0)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{aligneddefect}(1, 0)) = \mathrm{true} \wedge \mathrm{aligneddefect}(1, 2) = 93 \wedge \mathrm{vala}(\mathrm{aligneddefect}(1, 2)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{aligneddefect}(1, 2)) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(1, 2), 3) = \mathrm{true} \wedge \mathrm{aligneddefect}(4, 0) = 385 \wedge \mathrm{vala}(\mathrm{aligneddefect}(4, 0)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{aligneddefect}(4, 0)) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{aligneddefect}(4, 0), 4) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{seedvalcancellation}$. □

**Theorem 2.150.** *For every natural number $d$ and natural number $n$ and natural number $j$, if* $j < n$ *or* $n + 2 < j$, *then* $\mathrm{rowbit}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n)), n) = \mathrm{not}(\mathrm{rowbit}(d, n))$ *and* $\mathrm{rowbit}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n)), j) = \mathrm{rowbit}(d, j) \wedge \mathrm{vala}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n))) = \mathrm{vala}(d) \wedge \mathrm{valb}(\mathrm{wittxor}(d, \mathrm{centerflipmask}(n))) = \mathrm{valb}(d)$.

*Proof.* The goal follows from $\mathrm{centerflipmaskinvisible}(d, n, j)$. □

**Theorem 2.151.** *For every natural number $t$ and natural number $j$,* $\mathrm{rowbit}(\mathrm{observercode}(t), j) = \mathrm{leftcell}(t, j)$.

*Proof.* The goal follows from $\mathrm{observercodereadout}(t, j)$. □

**Theorem 2.152.** *For every natural number $Q$ and natural number $t$,* $\mathrm{observerdefect}(Q, t) = \mathrm{wittxor}(\mathrm{observercode}(t + Q), \mathrm{observercode}(t))$.

*Proof.* The goal follows from $\mathrm{observerdefectalignment}(Q, t)$. □

**Theorem 2.153.** *For every natural number $Q$ and natural number $t$ and natural number $j$,* $\mathrm{rowbit}(\mathrm{observerdefect}(Q, t), j) = \mathrm{rowbit}(\mathrm{aligneddefect}(Q, t), t + Q + j)$.

*Proof.* The goal follows from $\mathrm{observerdefectreadout}(Q, t, j)$. □

**Theorem 2.154.** *For every natural number $t$,* $\mathrm{vala}(\mathrm{observercode}(t + 1)) = \mathrm{bxor}(\mathrm{center}(t), \mathrm{bxor}(\mathrm{vala}(\mathrm{wittcarry}(\mathrm{observercode}(t))), \mathrm{observerinput}(t)))$ *and* $\mathrm{valb}(\mathrm{observercode}(t+1)) = \mathrm{bxor}(\mathrm{center}(t), \mathrm{valb}(\mathrm{wittcarry}(\mathrm{observercode}(t))))$.

*Proof.* The goal follows from $\mathrm{observervalstep}(t)$. □

**Theorem 2.155.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq t$*, then* $\mathrm{vala}(\mathrm{observerdefect}(m \cdot P, t+1)) = \mathrm{bxor}(\mathrm{vala}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t + m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(t)))), \mathrm{leftdefect}(m \cdot P, t, 1))$ *and* $\mathrm{valb}(\mathrm{observerdefect}(m \cdot P, t+1)) = \mathrm{valb}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{observercode}(t+m \cdot P)), \mathrm{wittcarry}(\mathrm{observercode}(t))))$.

*Proof.* The goal follows from $\mathrm{periodicobservervalstep}(T, P, m, t)$. □

**Theorem 2.156.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* $0 < P$*, then if* $0 < m$*, then if* $\mathrm{centerperiodic}(T, P)$*, then if* $T \leq t$*, then* $\mathrm{rowbit}(\mathrm{observerdefect}(m \cdot P, t), t + m \cdot P) = \mathrm{true}$ *and* $\mathrm{rowbit}(\mathrm{observerdefect}(m \cdot P, t), 0) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{periodicobserveredges}(T, P, m, t)$. □

**Theorem 2.157.** $\mathrm{observerdefect}(2, 0) = 7$ *and* $\mathrm{vala}(\mathrm{observerdefect}(2, 0)) = \mathrm{false} \wedge \mathrm{valb}(\mathrm{observerdefect}(2, 0)) = \mathrm{false} \wedge \mathrm{rowbit}(\mathrm{observerdefect}(2, 0), 0) = \mathrm{true} \wedge \mathrm{observerdefect}(4, 0) = 24 \wedge \mathrm{vala}(\mathrm{observerdefect}(4, 0)) = \mathrm{true} \wedge \mathrm{valb}(\mathrm{observerdefect}(4, 0)) = \mathrm{true} \wedge \mathrm{rowbit}(\mathrm{observerdefect}(4, 0), 0) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{observerseedzeroval}$. □

## 3 Cyclotomic observer transport

**Theorem 3.1.** *For every natural number $r$ and natural number $a$,* $\mathrm{qdivides}(r, \mathrm{wittlinpow}(r, a))$.

*Proof.* The goal follows from $\mathrm{qdividesintro}(r, a)$. □

**Theorem 3.2.** *For every natural number $r$ and natural number $d$, if* $\mathrm{qdivides}(r, d)$, *then there exists a natural number $a$ such that* $\mathrm{wittlinpow}(r, a) = d$.

*Proof.* The goal follows from $\mathrm{qdivideselim}(r, d)$. □

**Theorem 3.3.** *For every natural number $r$ and natural number $d$ and natural number $n$,* $\mathrm{qdivides}(r, \mathrm{wittshl}(d, n))$ *if and only if* $\mathrm{qdivides}(r, d)$.

*Proof.* The goal follows from $\mathrm{qdividesshl}(r, d, n)$. □

**Theorem 3.4.** *For every natural number $r$ and natural number $d$, if* $0 < r$, *then if* $\mathrm{qdivides}(r, d)$, *then* $\mathrm{vala}(d) = \mathrm{false}$ *and* $\mathrm{valb}(d) = \mathrm{false}$.

*Proof.* The goal follows from $\mathrm{qdividesval}(r, d)$. □

**Theorem 3.5.** *For every natural number $r$ and natural number $d$, if* $d \neq 0$, *then if* $\mathrm{qdivides}(r, d)$, *then* $2 \cdot r \leq \mathrm{binarydegree}(d)$.

*Proof.* The goal follows from $\mathrm{qdividesdegreebound}(r, d)$. □

**Theorem 3.6.** *For every natural number $d$, if* $d \neq 0$, *then there exists a natural number $r$ such that* $\mathrm{qdivides}(r, d)$ *and not* $\mathrm{qdivides}(r + 1, d)$.

*Proof.* The goal follows from $\mathrm{qorderexists}(d)$. □

**Theorem 3.7.** *For every natural number $r$ and natural number $d$, if* $d \neq 0$, *then* $\mathrm{qdivides}(r, d)$ *if and only if* $r \leq \mathrm{qorder}(d)$.

*Proof.* The goal follows from $\mathrm{qdividesifforder}(r, d)$. □

**Theorem 3.8.** *For every natural number $Q$ and natural number $t$,* $\mathrm{observersourceflux}(Q, t, 0) = 0$.

*Proof.* The goal follows from $\mathrm{observersourcefluxzero}(Q, t)$. □

**Theorem 3.9.** *For every natural number $Q$ and natural number $t$ and natural number $n$,* $\mathrm{observersourceflux}(Q, t, n{+}1) = \mathrm{wittxor}(\mathrm{wittlinear}(\mathrm{observersourceflux}(Q, t, n)), \mathrm{wittshl}(\mathrm{ob}$ $n), n + 1))$.

*Proof.* The goal follows from $\mathrm{observersourcefluxstep}(Q, t, n)$. □

**Theorem 3.10.** *For every natural number $Q$ and natural number $t$ and natural number $n$ and natural number $s$,* observersourceflux$(Q, t, n + s) =$ wittxor(wittlinpow$(s,$ observersourceflux$(Q, t, n)),$ wittshl(observersourceflux$(Q, t+ n, s), n)$).

*Proof.* The goal follows from observersourcefluxappend$(Q, t, n, s)$. □

**Theorem 3.11.** *For every natural number $Q$ and natural number $t$,* observerqrank$(Q, t) = t + Q - 2 \cdot$ qorder(observerdefect$(Q, t)$).

*Proof.* The goal follows from observerqrankdefinition$(Q, t)$. □

**Theorem 3.12.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then* wittshl(observerdefect$(m \cdot P, t+1), 1) =$ wittxor(wittlinear(observerdefect$(m \cdot P, t)),$ wittshl(observersource$(m \cdot P, t), 1)$).

*Proof.* The goal follows from periodicobserverpolynomialstep$(T, P, m, t)$. □

**Theorem 3.13.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then* observersource$(m \cdot P, t) =$ wittxor(wittxor(wittcarry(observercode$(t+ m \cdot P)),$ wittcarry(observercode$(t))),$ naturalbit(leftdefect$(m \cdot P, t, 1))$).

*Proof.* The goal follows from periodicobserversource$(T, P, m, t)$. □

**Theorem 3.14.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $n$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then* wittshl(observerdefect$(m \cdot P, t+n), n) =$ wittxor(wittlinpow$(n,$ observerdefect$(m \cdot P, t)),$ observersourceflux$(m \cdot P, t, n)$).

*Proof.* The goal follows from periodicobserverpolynomialwindow$(T, P, m, t, n)$. □

**Theorem 3.15.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $n$ and natural number $r$ and natural number $k$, if* centerperiodic$(T, P)$*, then if $T \leq t$, then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$*, then if $k \leq n+r$, then* qdivides$(k,$ observerdefect$(m \cdot P, t + n))$ *if and only if* qdivides$(k,$ observersourceflux$(m \cdot P, t, n))$.

*Proof.* The goal follows from periodicobserverwindowdivisibility$(T, P, m, t, n, r, k)$. □

**Theorem 3.16.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $r$, if* $\mathrm{centerperiodic}(T, P)$, *then if* $T \leq t$, *then if* $\mathrm{qdivides}(r, \mathrm{observerdefect}(m \cdot P, t))$, *then* $\mathrm{qdivides}(r + 1, \mathrm{observerdefect}(m \cdot P, t+1))$ *if and only if* $\mathrm{qdivides}(r+1, \mathrm{observersource}(m \cdot P, t))$.

*Proof.* The goal follows from $\mathrm{periodicobserverqgain}(T, P, m, t, r)$. □

**Theorem 3.17.** *For every natural number $Q$ and natural number $t$, if* $0 < Q$, *then* $2 \cdot \mathrm{qorder}(\mathrm{observerdefect}(Q, t)) \leq t + Q$.

*Proof.* The goal follows from $\mathrm{observerqorderbound}(Q, t)$. □

**Theorem 3.18.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $n$, if* $0 < P$, *then if* $0 < m$, *then if* $\mathrm{centerperiodic}(T, P)$, *then if* $T \leq t$, *then if* $0 < n$, *then* $\mathrm{observerqrank}(m \cdot P, t + n) < \mathrm{observerqrank}(m \cdot P, t)$ *if and only if* $\mathrm{qdivides}(\mathrm{qorder}(\mathrm{observerdefect}(m \cdot P, t)) + n/2 + 1, \mathrm{observersourceflux}(m \cdot P, t, n))$.

*Proof.* The goal follows from $\mathrm{periodicobserverqrankdrop}(T, P, m, t, n)$. □

**Theorem 3.19.** $\mathrm{observerdefect}(5, 2) = 216$ *and* $\mathrm{observerdefect}(5, 3) = 412 \wedge \mathrm{qdivides}(1, \mathrm{observerdefect}(5, 2)) \wedge \neg\, \mathrm{qdivides}(1, \mathrm{observerdefect}(5, 3)) \wedge \mathrm{center}(2) = \mathrm{false} \wedge \mathrm{center}(7) = \mathrm{false} \wedge \mathrm{center}(3) = \mathrm{true} \wedge \mathrm{center}(8) = \mathrm{true}$.

*Proof.* The goal follows from observerseedqloss. □

**Theorem 3.20.** *For every natural number $r$ and natural number $a$ and natural number $d$, if* $\mathrm{wittlinpow}(r, a) = d$, *then* $a \leq d$.

*Proof.* The goal follows from $\mathrm{qquotientbound}(r, a, d)$. □

**Theorem 3.21.** *For every natural number $d$ and natural number $r$, if* $\mathrm{qdivides}(r+1, d)$, *then* $\mathrm{qordersearch}(d, r + 1) = r + 1$.

*Proof.* The goal follows from $\mathrm{qordersearchhit}(d, r)$. □

**Theorem 3.22.** *For every natural number $d$ and natural number $r$, if not* $\mathrm{qdivides}(r + 1, d)$, *then* $\mathrm{qordersearch}(d, r + 1) = \mathrm{qordersearch}(d, r)$.

*Proof.* The goal follows from $\mathrm{qordersearchmiss}(d, r)$. □

**Theorem 3.23.** *For every natural number $d$,* $\mathrm{qorder}(d) = \mathrm{qordersearch}(d, \mathrm{binarydegree}(d))$.

*Proof.* The goal follows from qorderdefinition($d$). □

**Theorem 3.24.** *For every natural number $r$ and natural number $d$, if* qdivides($r, d$), *then there exists a natural number $a$ such that $a \leq d$ and* wittlinpow($r, a$) = $d$.

*Proof.* The goal follows from qdividesboundedwitness($r, d$). □

**Theorem 3.25.** *For every natural number $d$,* qordersearch($d, 0$) = 0.

*Proof.* The goal follows from qordersearchzero($d$). □

**Theorem 3.26.** *For every natural number $a$ and natural number $y$,* wittlinpow(twopow($a$), $y$) = wittxor(wittxor($y$, wittshl($y$, twopow($a$))), wittshl($y$, twopow($a$)+twopow($a$))).

*Proof.* The goal follows from qfrobeniustwopow($a, y$). □

**Theorem 3.27.** wittlinpow(16, 1) = wittxor(wittxor(1, wittshl(1, 16)), wittshl(1, 32)) *and* wittlinpow(16, wittxor(1, wittshl(1, 16))) = wittxor(1, wittshl(1, 48)).

*Proof.* The goal follows from qsixteenpage. □

**Theorem 3.28.** *For every natural number $a$ and natural number $y$,* qdivides(twopow($a$), wittxor($y$, witts twopow($a$)))).

*Proof.* The goal follows from qdividespageshifttwopow($a, y$). □

**Theorem 3.29.** *For every natural number $a$ and natural number $n$ and natural number $H$,* qdivides(twopow($a$), wittxor($H$, wittxor(pagefold(3·twopow($a$), $H, n$), pagetail(3· twopow($a$), $H, n$)))).

*Proof.* The goal follows from qdividespagefoldtwopow($a, n, H$). □

**Theorem 3.30.** *For every natural number $L$ and natural number $H$ and natural number $p$, if $0 < L$, then* page($L$, wittcarry($H$), $p$) = wittxor(wittcarry(page($L, H, p$)), naturalbit(pa

*Proof.* The goal follows from pagecarry($L, H, p$). □

**Theorem 3.31.** *For every natural number $L$ and natural number $H$ and natural number $n$, if $0 < L$, then* pagefold($L$, wittcarry($H$), $n$) = carryledger($L, H, n$).

*Proof.* The goal follows from carryfoldeqledger($L, H, n$). □

**Theorem 3.32.** *For every natural number $a$ and natural number $H$ and natural number $n$,* qdivides(twopow($a$), wittxor(wittcarry($H$), wittxor(carryledger(3· twopow($a$), $H, n$), pagetail(3 · twopow($a$), wittcarry($H$), $n$)))).

*Proof.* The goal follows from carryledgercongruencetwopow$(a, H, n)$. □

**Theorem 3.33.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $a$, if* centerperiodic$(T, P)$, *then if $T \leq t$, then* qdivides(twopow$(a)$, wittxor(observersource$(m \cdot P, t)$, wittxor(wittxor(carryledger(3·twopow$(a)$, observercode$(t+m \cdot P)$, $t+m \cdot P+1)$, carryledger(3·twopow$(a)$, observercode$(t)$, $t+m \cdot P+1)$), naturalbit(leftdefect$(m \cdot P, t, 1)$))))).

*Proof.* The goal follows from periodicobserversourceledger$(T, P, m, t, a)$. □

**Theorem 3.34.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$ and natural number $r$ and natural number $a$, if* centerperiodic$(T, P)$, *then if $T \leq t$, then if $r+1 \leq$* twopow$(a)$, *then if* qdivides$(r,$ observerdefect$(m \cdot P, t))$, *then* qdivides$(r+1,$ observerdefect$(m \cdot P, t+1))$ *if and only if* qdivides$(r+1,$ wittxor(wittxor(carryledger(3·twopow$(a)$, observercode$(t+m \cdot P)$, $t+m \cdot P+1)$, carryledger$(3 \cdot$ twopow$(a)$, observercode$(t)$, $t+m \cdot P+1)$), naturalbit(leftdefect$(m \cdot P, t, 1)$))).

*Proof.* The goal follows from periodicobserverqgainledger$(T, P, m, t, r, a)$. □

**Theorem 3.35.** qdivides(16, wittlinpow(16, 1)) *and* wittxor(wittphi(wittxor(2, wittlinpow(16, 1))), witt wittxor(wittlinpow(17, 1), 4)$\wedge\neg$ qdivides(16, wittxor(wittphi(wittxor(2, wittlinpow(16, 1))), wittphi(2)))

*Proof.* The goal follows from pageresiduenotclosed. □

**Theorem 3.36.** *For every natural number $a$ and natural number $A0$ and natural number $A1$ and natural number $A2$,* qdivides(twopow$(a)$, wittxor(wittxor(wittxor($A0$, wittshl($A1$ twopow$(a)$)), wittxor(wittxor($A0, A2$), wittshl(wittxor($A1, A2$), twopow$(a)$))))).

*Proof.* The goal follows from qblockresidue$(a, A0, A1, A2)$. □

**Theorem 3.37.** *For every natural number $a$ and natural number $A0$ and natural number $A1$ and natural number $A2$, if $A0 <$* twopow(twopow$(a)$), *then if $A1 <$* twopow(twopow$(a)$), *then if $A2 <$* twopow(twopow$(a)$), *then* qdivides(twopow$(a)$, wittxor(wittxor($A0$, wittshl($A1$, twopow$(a)$))), wittshl($A2$, twopow$(a)+$ twopow$(a)$))) *if and only if $A0 = A2$ and $A1 = A2$.*

*Proof.* The goal follows from qblockchargezeroiff$(a, A0, A1, A2)$. □

**Theorem 3.38.** *For every natural number $a$ and natural number $A$, if $A <$* twopow(twopow$(a)$), *then* wittcarry(wittxor(wittxor($A$, wittshl($A$, twopow$(a)$))), wittshl($A$, twopow$(a)+$ twopow$(a)$))) = wittxor(wittxor(wittxor(wittcarry($A$), wittshl(wittcarry($A$), twopow$(a)$))), wittshl(witt twopow$(a)$)), wittxor(wittshl(naturalbit(blockendpoint(twopow$(a), A)$), twopow$(a)$)), wittshl(naturalbit twopow$(a)$))).

*Proof.* The goal follows from repeatedblockcarry$(a, A)$. □

**Theorem 3.39.** *For every natural number $a$ and natural number $A$, if* $A <$ twopow(twopow$(a)$)*, then* qdivides(twopow$(a)$, wittxor(wittcarry(wittxor(wittxor($A$, wittshl($A$, twopow(twopow$(a)$)))), naturalbit(blockendpoint(twopow$(a), A$)))).

*Proof.* The goal follows from repeatedblockcarrycongruence$(a, A)$. □

**Theorem 3.40.** *For every natural number $a$ and natural number $H$ and natural number $p$ and natural number $A$, if* $A <$ twopow(twopow$(a)$)*, then if* page(3·twopow$(a), H, p)$ = wittxor(wittxor($A$, wittshl($A$, twopow$(a)$)), wittshl($A$, twopow$(a)$+ twopow$(a)$))*, then* qdivides(twopow$(a)$, wittxor(page(3·twopow$(a)$, wittcarry$(H), p)$, wittxor(naturalbit twopow$(a), H, p)$))))).

*Proof.* The goal follows from pagecarryrepeated$(a, H, p, A)$. □

**Theorem 3.41.** *For every natural number $T$ and natural number $P$ and natural number $m$ and natural number $t$, if* centerperiodic$(T, P)$*, then if* $T \leq t$*, then* observersource$(m \cdot P, t)$ = wittxor(wittxor(wittcarry(observerdefect($m \cdot P, t$)), crosscarry(observercode$(t)$, observerdefect($m \cdot P, t$))), naturalbit(leftdefect($m \cdot P, t, 1$))).

*Proof.* The goal follows from periodicobserversourcecrosscarry$(T, P, m, t)$. □

**Theorem 3.42.** *There exists a natural number $s$ such that there exists a natural number $s2$ such that* qdivides(16, wittxor$(s, s2)$) *and not* qdivides(16, wittxor(wittphi$(s)$, wittphi$(s2$)

*Proof.* The goal follows from residuenotcongruence. □

**Theorem 3.43.** *For every natural number $Q$ and natural number $t$ and natural number $j$,* rowbit(observerdefect$(Q, t), j)$ = leftdefect$(Q, t, j)$.

*Proof.* The goal follows from observerdefectleftreadout$(Q, t, j)$. □

**Theorem 3.44.** *For every natural number $r$ and natural number $e$ and natural number $k$, if* $e \neq 0$*, then if* qdivides$(r,$ wittshl$(e, k))$*, then* $k + 2 \cdot r \leq$ binarydegree(wittshl$(e, k)$).

*Proof.* The goal follows from qfrontdegreebudget$(r, e, k)$. □

**Theorem 3.45.** *For every natural number $n$ and natural number $e$ and natural number $o$, if* $e \neq 0$*, then if* twopow$(n) \leq o$*, then if* wittshl$(e, o) <$ twopow$(3 \cdot$ twopow$(n))$*, then not* qdivides(twopow$(n)$, wittshl$(e, o)$).

*Proof.* The goal follows from qpageoffsetcharge$(n, e, o)$. □

**Theorem 3.46.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < Q$, then if* rightmostleft$(Q, t, k)$*, then* $k + 2 \cdot$ qorder(observerdefect$(Q, t)) \leq t + Q$.

*Proof.* The goal follows from observerfrontqbudget$(Q, t, k)$. □

**Theorem 3.47.** *For every natural number $Q$ and natural number $t$ and natural number $k$, if $0 < Q$, then if* rightmostleft$(Q, t, k)$*, then* $k \leq$ observerqrank$(Q, t)$.

*Proof.* The goal follows from observerfrontqrank$(Q, t, k)$. □

**Theorem 3.48.** *For every natural number $a$ and natural number $Q$ and natural number $t$ and natural number $p$ and natural number $o$, if* twopow$(a) \leq o$*, then if $o < 3 \cdot$* twopow$(a)$*, then if* rightmostleft$(Q, t, 3 \cdot$ twopow$(a) \cdot p + o)$*, then not* qdivides(twopow$(a)$, page$(3 \cdot$ twopow$(a)$, observerdefect$(Q, t), p))$.

*Proof.* The goal follows from observerfrontpagecharge$(a, Q, t, p, o)$. □